

Homotopy Continuation in Economics

Kevin Phan (Under Dr. Max Hill)

Department of Mathematics, University of Hawai'i at Mānoa



What is Homotopy Continuation?

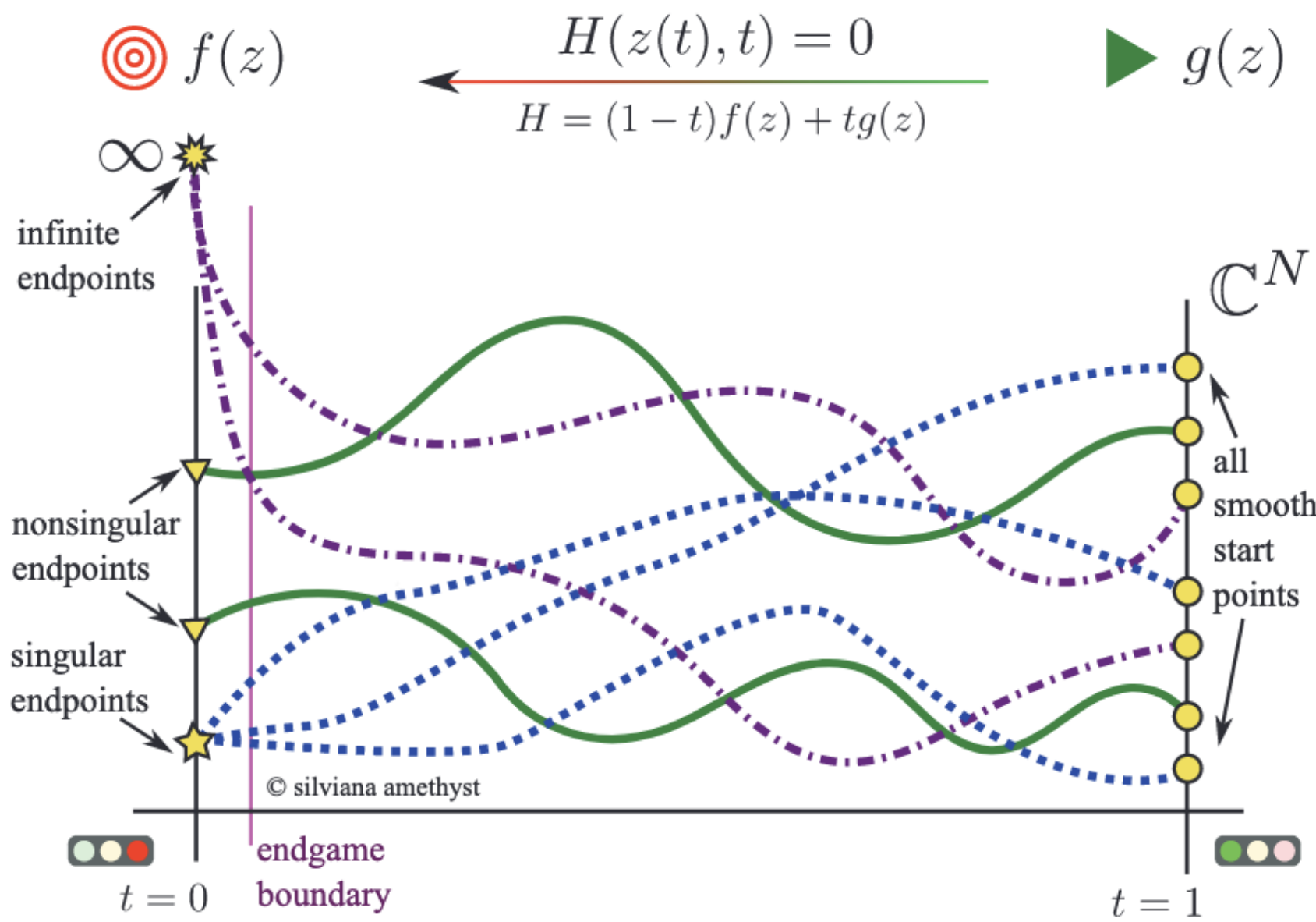


Figure 1. Illustrative Example from Duff & Regan

Homotopy Continuation is a numerical method that leverages concepts from algebraic geometry to solve nonlinear systems of polynomial equations. There are many examples ranging from Physics, Robotics, Ecology, and in this poster, Economics. We will use the programming language Julia with the library HomotopyContinuation.jl.

Background

A homotopy is a continuous function H between two continuous functions f, g from the topological spaces X and Y such that

$$H : X \times [0, 1] \rightarrow Y$$

such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$.

- A system of equations $g(\mathbf{z}) = 0$ where the solutions are known is called the **start system**.
- The system of equations that we want to eventually reach (with continuous deformation) is the **target system**, another system of equations $f(\mathbf{z}) = 0$, whose solutions we want to know.

There are several types of homotopies and ways to compute solutions. For now, we will focus on the **Total Degree Homotopy** that uses an intuitive start system.

Example of Total Degree Homotopy

$$f(x, y) = \begin{pmatrix} x^2 + y^2 - 3 \\ 2x^2 + 0.5x * y + 3y^2 - 2 \end{pmatrix}$$

And the corresponding total degree start system.

$$g(x, y) = \begin{pmatrix} x^2 - 1 \\ y^2 - 1 \end{pmatrix}$$

We have (1,1), (-1,1), (1,-1), (-1,-1) as starting solutions.

Example Results

Fortunately, this example is quite simple. There are four non-singular paths that are real. However, the solutions are actually complex.

Solution	x Coordinate	y Coordinate
1	$2.47 - 0.42i$	$-0.58 - 1.80i$
2	$-2.47 - 0.42i$	$0.58 - 1.80i$
3	$2.47 + 0.42i$	$-0.58 + 1.80i$
4	$-2.47 + 0.42i$	$0.58 + 1.80i$

Table 1. Complex Solutions for the Initial System f

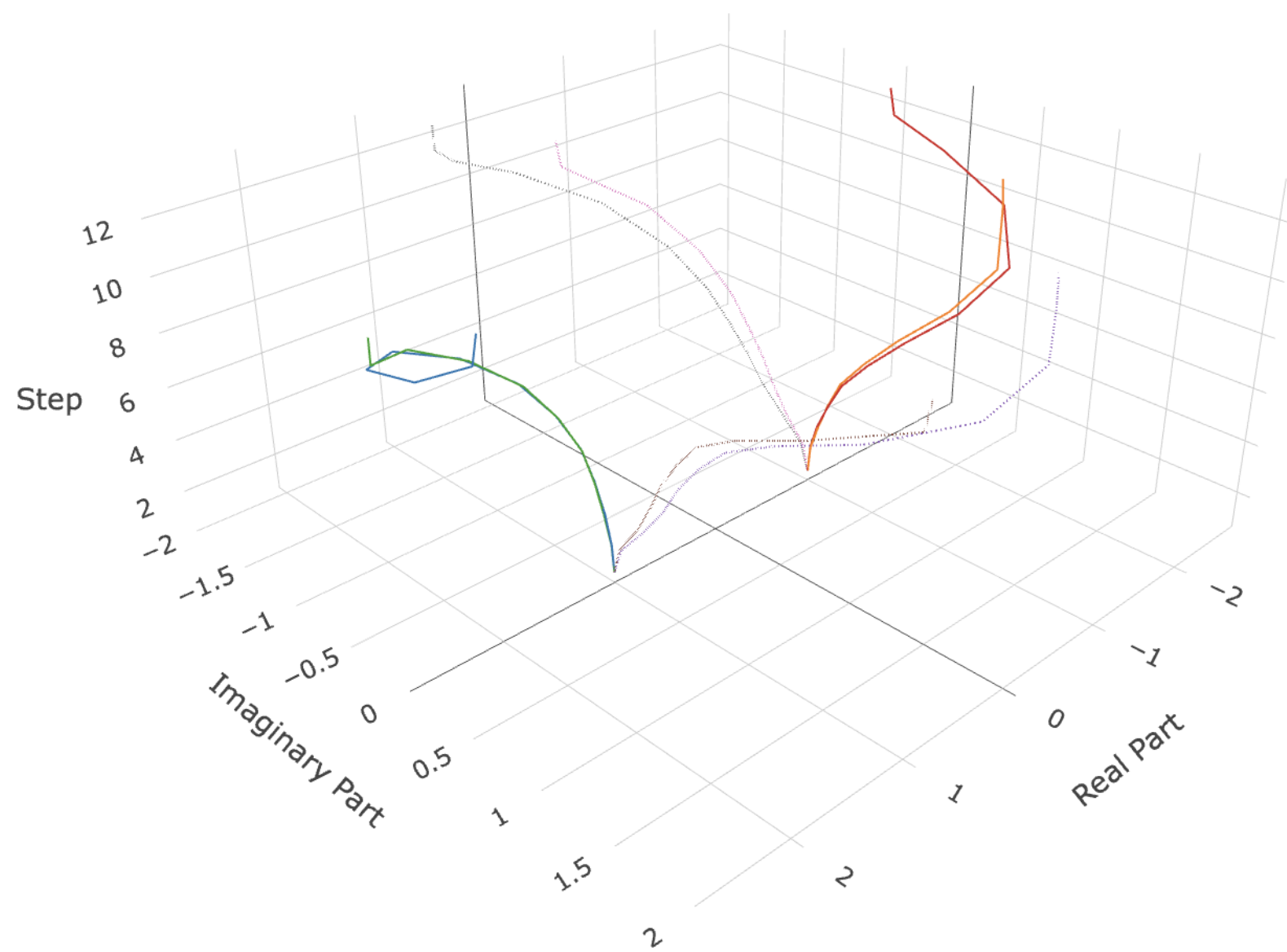


Figure 2. Simple Example Visualized (Bold: x, Dotted: y)

The Bertrand Model

Given two firms (a duopoly) that are subject to different consumers/products (and consequently different demand curves), they attempt to find a pure-strategy Nash equilibrium where they maximize profit.

We will consider an example that was created into a systems of polynomial equations below, where Z was constructed as a trick to avoid a rational expression. f_1 and f_2 were constructed using equations that are omitted from this poster.

$$\begin{aligned} f_1 &= -2700 + 2700p_x + 8100Z^2p_x^2 - 5400Z^2p_x^3 + 51Z^3p_x^6 - 2Z^3p_x^7 \\ f_2 &= -2700 + 2700 * p_y + 8100Z^2p_y^2 - 5400Z^2 * p_y^3 + 51Z^3p_y^6 - 2Z^3p_y^7 \\ Z &= Z^2p_x^2p_y^2 - (p_x^2 + p_y^2) \end{aligned}$$

Here, we also use a total degree homotopy, where the solutions are the unitary roots. We use Bézout's Theorem to see that there should be an upper bound of 600 solutions/paths.

Results of Bertrand Model with Homotopy Continuation

In the graph below, there are 600 solutions that are generated, but many of them veer of into infinity and are not relevant. Additionally, we also use some first and second order conditions to narrow down our solutions. For example, negative or infinite solutions are not valid for Economics.

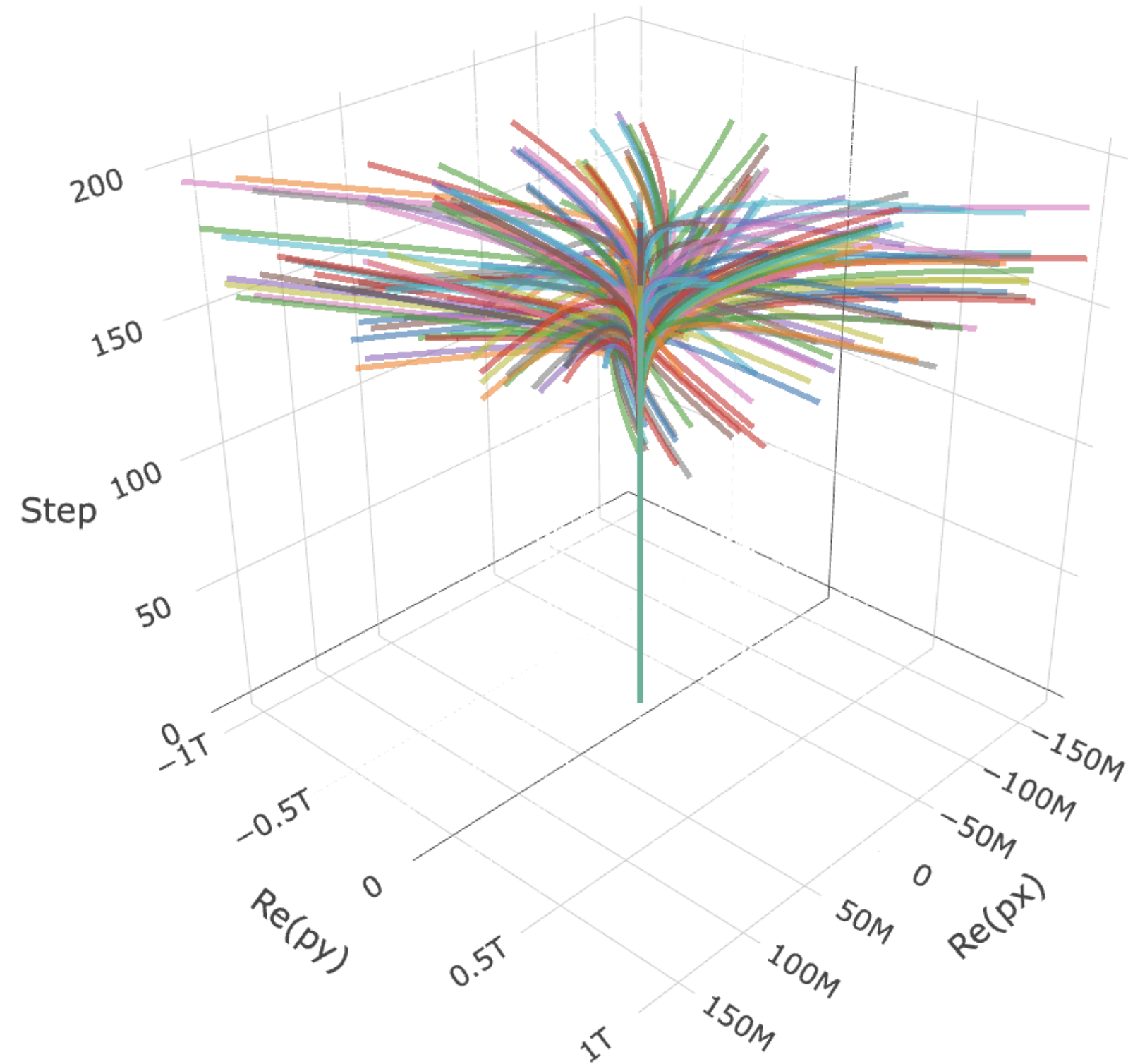


Figure 3. Results of Bertrand Model with Homotopy Continuation

Narrowing it down to non-singular solutions we get 62, 18 real. But there are only 4 relevant solutions.

Equilibrium Prices

$$\begin{aligned} (p_x^1, p_y^1) &= (1.757, 1.757) & (p_x^2, p_y^2) &= (22.987, 22.987) \\ (p_x^3, p_y^3) &= (2.168, 25.157) & (p_x^4, p_y^4) &= (25.157, 2.168) \end{aligned}$$

Table 2. Relevant Strategies for the Bertrand Model

References

- [1] T. Duff and M. Regan. Polynomial Systems, Homotopy Continuation, and Applications. *Notices of the American Mathematical Society*, 70(01):1, Jan. 2023.
- [2] P. J. Renner. *Applications of algebraic geometry in economics*. PhD thesis, ETH Zurich, 2013. Artwork Size: 101 p. Medium: application/pdf Pages: 101 p.